

Towards development of evidence to inform recommendations for the evaluation and management of bronchiectasis.

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Bronchiectasis (BE) is a chronic condition characterized by airway dilation as a consequence of a variety of pathogenic processes. It is often associated with persistent airway infection and an inflammatory response resulting in cough productive of purulent sputum, which has an adverse impact on quality of life. The prevalence of BE is increasing worldwide. Treatment guidelines exist for managing BE, but they are generally informed by a paucity of high-quality evidence. This review presents the findings of a scientific advisory board of experts held in the United States in November 2020. The main focus of the meeting was to identify unmet needs in BE and propose ways to identify research priorities for the management of BE, with a view to developing evidence-based treatment recommendations. Key issues identified include diagnosis, patient evaluation, promoting airway clearance and appropriate use of antimicrobials. Unmet needs include effective pharmacological agents to promote airway clearance and reduce inflammation, control of chronic infection, clinical endpoints to be used in the design of BE clinical trials, and more accurate classification of patients using phenotypes and endotypes to better guide treatment decisions and improve outcomes.

Bronchiectasis in African children: Challenges and barriers to care.

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Bronchiectasis (BE) is a chronic condition affecting the bronchial tree. It is characterized by the dilatation of large and medium-sized airways, secondary to damage of the underlying bronchial wall structural elements and accompanied by the clinical picture of recurrent or persistent cough. Despite an increased awareness of childhood BE, there is still a paucity of data on the epidemiology, pathophysiological phenotypes, diagnosis, management, and outcomes in Africa where the prevalence is mostly unmeasured, and likely to be higher than high-income countries. Diagnostic pathways and management principles have largely been extrapolated from approaches in adults and children in high-income countries or from data in children with cystic fibrosis. Here we provide an overview of pediatric BE in Africa, highlighting risk factors, diagnostic and management challenges, need for a global approach to addressing key research gaps, and recommendations for practitioners working in Africa.

Aspergillus-Associated Endophenotypes in Bronchiectasis.

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Bronchiectasis is a chronic condition of global relevance resulting in permanent and irreversible structural airway damage. Bacterial infection in bronchiectasis is well studied; however, recent molecular studies identify fungi as important pathogens, either independently or in association with bacteria. *Aspergillus* species are established fungal pathogens in cystic fibrosis and their role is now increasingly being recognized in noncystic fibrosis bronchiectasis. While the healthy airway is constantly exposed to ubiquitously present *Aspergillus* conidia in the environment, anatomically damaged airways appear more prone to colonization and subsequent infection by this fungal group. *Aspergilli* possess diverse immunopathological mechanistic capabilities and when coupled with innate immune defects in a

susceptible host, such as that observed in bronchiectasis, it may promote a range of clinical manifestations including sensitization, allergic bronchopulmonary aspergillosis, *Aspergillus* bronchitis, and/or invasive aspergillosis. How such clinical states influence "endophenotypes" in bronchiectasis is therefore of importance, as each *Aspergillus*-associated disease state has overlapping features with bronchiectasis itself, and can evolve, depending on underlying host immunity from one type into another. Concurrent *Aspergillus* infection complicates the clinical course and exacerbations in bronchiectasis and therefore dedicated research to better understand the *Aspergillus*-host interaction in the bronchiectasis airway is now warranted.

Bronchiectasis-Could Immunonutrition Have a Role to Play in Future Management?

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Bronchiectasis is a chronic condition in which areas of the bronchial tubes become permanently widened predisposing the lungs to infection. Bronchiectasis is an age-associated disease with the highest prevalence in people older than 75 years. While the prevalence of bronchiectasis is higher in males, disease is more severe in females who have a poorer prognosis. The overall prevalence of the disease is thought to be rising. Its aetiology is multi-faceted, but a compromised immune system is now thought to play a central role in the pathology of this disease. Research has begun to study the role of malnutrition and certain nutrients-vitamin D and zinc-along with the role of the lung microbiome in relation to the management of bronchiectasis. Given this, the present mini review sets out to provide an overview of the state-of-the-art within the field, identify research gaps and pave the way for future developments and research investment within this field.

What's new in the management of adult bronchiectasis?

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Bronchiectasis is a heterogeneous, chronic condition with many aetiologies. It poses a significant burden on patients and healthcare practitioners and services. Clinical exacerbations often result in reduced quality of life, increased rate of lung function decline, increased hospitalisation, and mortality. Recent focus in respiratory research, guidelines, and future management options has improved this clinical field in evidence-based practice, but further work and phase III clinical trials are required. This article aims to summarise and explore advances in management strategies in recent years and highlight areas of research and future focus.

Paediatric and adult bronchiectasis: Monitoring, cross-infection, role of multidisciplinary teams and self-management plans.

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Bronchiectasis is a chronic lung disease associated with structurally abnormal bronchi, clinically manifested by a persistent wet/productive cough, airway infections and recurrent exacerbations. Early identification and treatment of acute exacerbations is an integral part of monitoring and annual review,

in both adults and children, to minimize further damage due to infection and inflammation. Common modalities used to monitor disease progression include clinical signs and symptoms, frequency of exacerbations and/or number of hospital admissions, lung function (forced expiratory volume in 1 s (FEV1) % predicted), imaging (radiological severity of disease) and sputum microbiology (chronic infection with *Pseudomonas aeruginosa*). There is good evidence that these monitoring tools can be used to accurately assess severity of disease and predict prognosis in terms of mortality and future hospitalization. Other tools that are currently used in research settings such as health-related quality of life (QoL) questionnaires, magnetic resonance imaging and lung clearance index can be burdensome and require additional expertise or resource, which limits their use in clinical practice. Studies have demonstrated that cross-infection, especially with *P. aeruginosa* between patients with bronchiectasis is possible but infrequent. This should not limit participation of patients in group activities such as pulmonary rehabilitation, and simple infection control measures should be carried out to limit the risk of cross-transmission. A multidisciplinary approach to care which includes respiratory physicians, chest physiotherapists, nurse specialists and other allied health professionals are vital in providing holistic care. Patient education and personalized self-management plans are also important despite limited evidence it improves QoL or frequency of exacerbations.

Medical nutritional therapy in infected bronchiectasis with impending respiratory failure: A case report.

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Bronchiectasis is a chronic infective and inflammatory respiratory disease that causes significant morbidity and mortality. Repeated respiratory infections may lead to infected bronchiectasis (IB) and acute exacerbations which often require hospital admission, increase risk of malnutrition and impact quality of life and eventually leads to death. Nutritional therapy is needed to modulate inflammation and enhance immunity to reduce severity of exacerbation, overcome malnutrition, as well as to decrease morbidity and mortality. A 59-year-old female patient, diagnosed with IB. The patient had low oral intake due to productive cough and anorexia since 2 weeks before admission. Moreover, she had gradual shortness of breath that caused an impending respiratory failure during hospitalization, supported by continuous positive airway pressure (CPAP). Nutritional assessment was made based on Subjective Global Assessment (SGA) score C. Abnormal laboratory findings seen were increased in neutrophil-to-lymphocyte ratio (NLR) 9.3, moderate depletion of immune system with total lymphocyte count (TLC) 808.4/ μ l, hypoalbuminemia (3.2g/dl) and increased in liver enzymes: aspartate aminotransferase (AST) 206U/l, Alanine aminotransferase (ALT) 224U/l. Nutritional therapy was given gradually with target calorie 1400-1900kcal, protein 0.8-1.5g/kg IBW/day, carbohydrates 45-50%, and fat 33.3-43% through oral and parenteral nutrition. The patient was given supplementations such as vitamins (A, B complex, C, D), zinc, curcumin and snakehead fish extract high albumin content. After 14 days of treatment, significant clinical and metabolic improvement in NLR, TLC, plasma albumin, liver enzymes (AST/ALT), blood gas analysis, and functional capacity (handgrip strength) were found. An adequate nutritional therapy with macro and micro-nutrients in IB patient can improve clinical outcome, nutritional status and quality of life.

Paediatric and adult bronchiectasis: Diagnosis, disease burden and prognosis.

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Bronchiectasis is a chronic, debilitating disease with increasing worldwide prevalence and burden. Accurate and early diagnosis is essential for both its management and prognosis. This review will discuss the diagnosis of bronchiectasis, the international burden of the disease and its current prognosis.

Ciprofloxacin DPI in non-cystic fibrosis bronchiectasis: a Phase II randomized study.

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In non-cystic fibrosis (CF) bronchiectasis, the chronic airways infection is a risk factor for frequent infectious exacerbations and for an overall poor disease prognosis. Its therapy with inhaled antibiotics might be useful alike in the case of the CF-related disease. Analysis and discussion of the results of a Phase II study evaluating the efficacy and safety of the inhaled ciprofloxacin-dry powder (C-DPI) formulation in patients with non-CF bronchiectasis. On short-term basis C-DPI is able to reduce the sputum bacterial load in such patients but long-term studies are also needed to better characterize the clinical efficacy of this compound.

Developing drug therapies in bronchiectasis.

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Bronchiectasis is a chronic respiratory condition characterised by cough, sputum production and recurrent chest infections. There are multiple aetiologies; but in up to 50% of patients, the aetiology is unknown. The treatment is largely symptomatic with regular chest physiotherapy and antibiotics for infective exacerbations. Research is being directed towards breaking the 'vicious circle' of bronchiectasis with therapies directed at improving mucociliary clearance, treating chronic infection and reducing inflammation in the airways. This review highlights the current status of bronchiectasis research, summarising reported and ongoing studies of potential therapeutic agents not yet assessed in large trials or licensed for treatment. A literature review was performed using the PubMed database and upcoming trials were sought on the ClinicalTrials.gov website. The article is limited to studies in preclinical to Phase II clinical trials. The trials highlighted in this article offer insight into potential therapeutic agents for the future and help highlight areas in need of further targeted research. There are promising new anti-infective and anti-inflammatory therapies for more advanced bronchiectasis. That being said, Phase III studies are still needed to investigate these agents further, as well as at what stage therapy should be implemented.
